SUPPORTING ADULT GUIDANCE

Adding and subtracting algebraic fractions

Algebraic Expressions and Identities · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson carries the number method for adding fractions over to algebra. In the main activity, learners choose a multiplier for each fraction card; the cards rewrite themselves and turn green when the denominators match (a larger-than-necessary common denominator is nudged). They then combine the numerators — with a named check for the sign slip when subtracting — and simplify, with a table of values confirming the answer. They spot slips in worked calculations and, in Creativity, split a fraction into two (a first glimpse of partial fractions).

Driving Question: *How do fractions with letters share a denominator?*

The mathematics, in plain terms

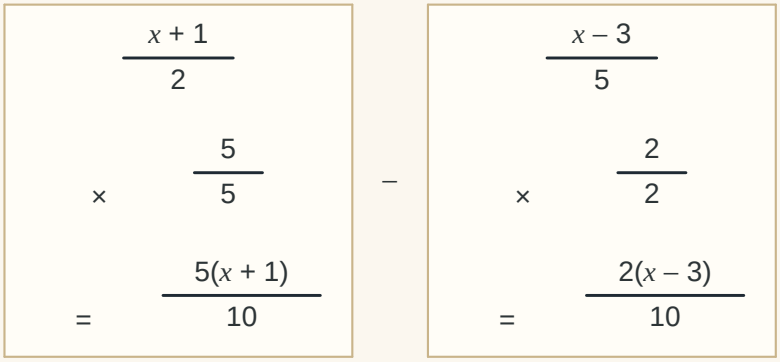
$a/b + c/d = (ad + bc)/bd$, since multiplying by d/d and b/b multiplies by 1. When b and d share a factor, the lowest common denominator (their LCM) keeps expressions smaller. In subtraction the second numerator is negated as a whole. The result should be simplified and is valid wherever the original denominators are non-zero.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Subtracting a whole numerator

Work out $(x + 1)/2 - (x - 3)/5$.

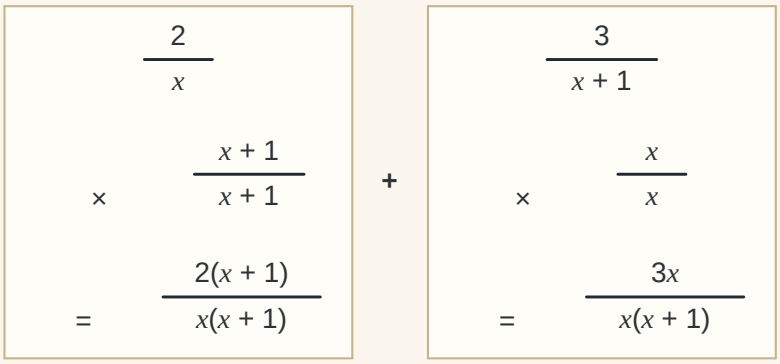


1. Both cards now have denominator 10: each was multiplied by a form of 1.	5/5 and 2/2
2. Combine the numerators, subtracting the whole second one.	$5(x + 1) - 2(x - 3)$
3. Expand: the –2 changes every sign in $(x - 3)$.	$5x + 5 - 2x + 6$
4. Collect.	$3x + 11$

$(x + 1)/2 - (x - 3)/5 = (3x + 11)/10$.

Example 2 — Letters in the denominators

Work out $\frac{2}{x} + \frac{3}{x+1}$.



1. The common denominator is $x(x+1)$: each card gets what it is missing.	$\times (x+1)/(x+1)$ and $\times x/x$
2. Add the numerators.	$2(x+1) + 3x = 5x + 2$
3. Check with $x = 1$: $2/1 + 3/2 = 3.5$, and $(5+2)/(1 \times 2) = 3.5$.	✓

$\frac{2}{x} + \frac{3}{x+1} = \frac{5x+2}{x(x+1)}$.

The number method, unchanged

Nothing new is introduced: the warm-up is number fractions, and the builder makes 'multiply by a form of 1' visible on each card. The table of values ties back to Lesson 1's checking and Lesson 4's counterexamples. The stretch proves the general rule and shows why $(a + c)/(b + d)$ is wrong.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Fractions you already know: 2/3 + 1/4, and why it is not 3/7.	~5 min
Movement	The common-denominator builder: cards, combine, simplify, check. Stretch: prove $a/b + c/d = (ad + bc)/bd$.	~18 min
Reflection	Spot the slip in worked calculations — or decide every line is right.	~8 min
Creativity	Split it apart: one fraction into two.	~9 min
Rest	Multiplying by $(x+1)/(x+1)$ is multiplying by 1.	~2 min
Nutrition	Intellectual: transfer — trusting a number method with letters.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding numerators and denominators: 2/3 + 1/4 = 3/7.
Ask: “Is 3/7 bigger or smaller than 2/3? Can adding make it smaller?”

Multiplying only the denominator.
Ask: “Is the value still the same? What did you multiply the fraction by?”

Sign slip: $-(x - 3)$ written as $-x - 3$.
Ask: “What is –1 × –3?”

Using a common denominator that is bigger than needed.
Ask: “Do the denominators share a factor?”

Forgetting to simplify at the end.
Ask: “Does the numerator share a factor with the denominator?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: adding number fractions, expanding brackets, and factorising. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner gets $(x + 6)/6$ for $(x + 4)/3 - (x - 2)/6$. The aim is a question that lets them find the slip.

- “Try $x = 0$ in the original and in your answer. Do they agree?”
- “What is $2(x + 4) - (x - 2)$ when you expand it?”

- “What does the minus sign do to the –2?”
- “What should the numerator be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Must learners use the lowest common denominator?
Any common denominator gives a correct answer, but the lowest keeps the algebra smaller. The lesson nudges towards it.

Is a factorised or expanded denominator accepted?
Both: $(x - 1)(x + 2)$ and $x^2 + x - 2$ are equally acceptable.

What is 'Split it apart'?
Writing one fraction as a sum of two simpler ones — called partial fractions at A level. Here it is just the method run backwards.

What comes next?
Lesson 6: changing the subject when it appears twice, or as a square.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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